FRAMEWORK FOR AI USE CASE INTEGRATION IN CURRICULUM DELIVERY

Part A: AI Use Case Profile

1. **Use Case ID:** KJS-AGR-01
2. **Use Case Title:** Irrigation Advisory System for Sugarcane Crop using AI and Sensor-based Technology
3. **AI Application Vertical:** Agriculture
4. **Collaborating Organization:** K J Somaiya Applied Agricultural Research (KIAAR), Sameerwadi, Karnataka, Godavari Biorefineries Ltd. (GBL)
5. **Beneficiaries:** Farmers (approx. 18,000–25,000 sugarcane farmers associated with GBL and KIAAR, Cane department supervisors and field assistants, Agricultural researchers and agronomists, Sugar factory management for planning cane availability and productivity.
6. **Problem Statement:** Sugarcane cultivation requires large quantities of water, and irrigation scheduling is generally based on farmers' experience rather than scientific recommendations. Due to inadequate knowledge of crop water requirements, soil moisture conditions, weather forecasts, and irrigation timing, farmers often practice over-irrigation or delayed irrigation, resulting in:

   - Water wastage
   - Reduced crop productivity
   - Yield loss due to moisture stress
   - Increased energy consumption
   - Higher cultivation costs
   - Nutrient losses because of improper fertigation
   - Inefficient use of electricity supplied for irrigation pumps
   - Difficulty in monitoring thousands of spatially distributed farms

   There is a need for an AI-driven irrigation advisory system that integrates field sensors, weather forecasts, geospatial information, and machine learning models to provide plot-specific irrigation recommendations and decision support to farmers in a simple and understandable manner.

7. **Background and Context:**

   The KIAAR and GBL ecosystem supports nearly 18,000–25,000 sugarcane farmers spread across Northern Karnataka. Through previous geofencing initiatives, thousands of sugarcane plots have already been digitized and spatially mapped. Soil testing records, farmer details, and crop information are available through existing digital platforms. However, irrigation decisions are still largely manual and depend on experience rather than data-driven approaches. Variations in soil types, crop stages, weather conditions, rainfall events, and water availability necessitate dynamic irrigation recommendations.

   Recent advances in IoT and sensor technologies, Remote sensing, Artificial Intelligence and Machine Learning, Weather forecasting systems, Geospatial analytics, Mobile applications and dashboards, etc. provide an opportunity to build a smart irrigation advisory system capable of delivering personalized recommendations at the plot level.

   The proposed system aims to combine sensor observations, satellite-derived information, weather forecasts, and AI models to improve irrigation efficiency, enhance water productivity, and promote sustainable sugarcane cultivation.

8. Objectives:

1) Develop an AI-enabled irrigation advisory system for sugarcane cultivation that integrates field observations, weather information, and crop-specific requirements to support informed irrigation decision-making.
2) Monitor soil moisture, environmental conditions, and crop status using sensor networks and geospatial technologies to enable plot-level assessment and precision irrigation management.
3) Generate real-time, forecast-based, and plot-specific irrigation and fertigation recommendations to optimize irrigation scheduling, improve resource utilization, and support operational constraints such as electricity availability.
4) Develop farmer-friendly web and mobile platforms for advisory dissemination, visualization of geofenced plots, crop information management, and decision support.
5) Enhance water use efficiency, reduce yield losses due to water stress, and improve the sustainability and climate resilience of sugarcane farming practices.

9. Solution Approach:

The proposed system will integrate multiple data sources and AI models to provide intelligent irrigation recommendations: IoT Sensors, Geospatial Data, Weather Data, and Farm Information and develop the below listed AI Models for Prediction:

- Next Irrigation Date Prediction
- Irrigation Duration Prediction
- Crop Water Requirement Prediction
- Water Stress Probability Model
- Rainfall-Adjusted Irrigation Recommendation
- Yield Loss Prediction Due to Delayed Irrigation
- Pump Scheduling Optimization
- Fertigation Recommendation Model
- Disease and Water Stress Forecasting
- Yield Prediction Model
- Farmer-Friendly Advisory Generation Model (LLM-Based)

10. Operational Challenges:

A. Sensor-related

- Sensor calibration and maintenance.
- Power supply and battery life.
- Data communication failures.
- Missing or noisy sensor data.

B. AI Challenges

- Limited labeled datasets.
- Model generalization across soil types.
- Seasonal variability.

C. Geospatial Challenges

- Accurate plot mapping.
- Dynamic crop boundaries.
- Multi-source data integration.

D. Weather Challenges

- Reliability of rainfall forecasts.
- Micro-climatic variations.

E. Implementation Challenges

- Connectivity issues in villages.
- Farmer adoption and trust.
- Device maintenance.
- Training requirements.
- Scalability to 25,000 farms.
- Continuous model updating.
- Integration with existing STEPS (Somaiya Technology for Efficient Plot Survey) platforms.

11. High-Level Workflow:

1) Data Acquisition: Collection of field, crop, soil, weather, and geospatial information through mobile applications, databases, APIs, sensor networks, IoT devices, satellite imagery, and GIS-based plot digitization.
2) Data Integration: Integration of sensor observations, weather data, satellite imagery, soil reports, farmer records, and historical yield information into a unified agricultural data platform.
3) AI Analytics: Development of AI/ML models for soil moisture forecasting, irrigation scheduling, water stress prediction, fertigation planning, pump scheduling optimization, yield loss estimation, and yield prediction.
4) Decision Support: Generation of plot-specific irrigation and fertigation recommendations based on field conditions, crop requirements, weather information, and predictive analytics.
5) Dashboard and Advisory Platform: Delivery of recommendations through web and mobile applications featuring farmer dashboards, spatial visualization of plots, irrigation and weather information, soil and crop health monitoring, fertigation guidance, yield analytics, and alert generation.

Details of workflow activities:

Workflow Stage	Activities	Details
1. Field Data Acquisition	Sensor deployment, IoT devices, data collection	Design sensor interfaces, communication modules, and wireless data acquisition systems (EXTC)
		Development of embedded systems and edge devices for sensor integration (ECE)
		Communication protocols and IoT gateway connectivity (CCE)
		Smart sensing systems and autonomous monitoring solutions (RAI)
2. Data Transmission and Integration	Data ingestion, cloud connectivity, database creation	Development of APIs, cloud infrastructure, database management and integration platforms (IT)
		Backend systems, middleware, and data synchronization mechanisms (CE)
		Enterprise applications and process workflows for agricultural data management (CSBS)
3. Data Pre-processing and Feature Engineering	Data cleaning, normalization, feature extraction	Data pre-processing, feature engineering, and preparation of datasets for machine learning models (IT, CE, AI&DS)
		Development of data pipelines and automated processing frameworks (IT, CE)
4. AI Analytics and Predictive	Prediction of irrigation date,	Development and validation of machine learning and deep learning models (IT, CE, AI&DS)

Modeling	duration, water stress, yield loss, pump scheduling	Intelligent decision-making algorithms and optimization models (RAI)
		AI model deployment and inference frameworks (CE)
5. Decision Support Engine	Rule-based recommendations and optimization	Design of business logic and recommendation engines (IT)
		Workflow optimization and integration with enterprise applications (CSBS)
		AI-assisted decision-making and predictive analytics (IT, CE, AI&DS)
6. Generative AI Advisory Layer	Conversion of predictions into farmer-friendly advisories	Large Language Models (LLMs), multilingual advisory generation, chatbot development (IT, CE, AI&DS)
		Development of AI service integration and API interfaces (IT, CE)
		Integration of AI models into applications and scalable deployment (CE)
7. Dashboard and Mobile Application Development	Web and mobile interfaces, visualization, GIS maps	Full-stack web and mobile application development (IT, CE)
		Backend services, APIs, authentication, and database integration (CE)
		User workflows, reporting systems, and management dashboards (CSBS)
		AI visualization and analytics dashboards (IT, CE, AI&DS)
8. Smart Irrigation Automation	Automated pump operation and scheduling	Embedded controllers and smart irrigation hardware (ECE)
		Wireless communication and control systems (EXTC)
		Intelligent autonomous irrigation control systems (RAI)
		IoT communication protocols and remote monitoring (CCE)
9. Farmer Interaction and Support Systems	Notifications, chatbot, advisory dissemination	Mobile applications and notification services (IT, CE)
		Conversational AI and personalized recommendations (IT, CE, AI&DS)
		User experience and service workflows (CSBS)
10. Continuous Learning and System Improvement	Feedback collection, model retraining, performance monitoring	Continuous learning frameworks and model enhancement (IT, CE)
		Monitoring platforms and cloud services (IT, CE)
		System scalability and performance optimization (IT, CE)
		Adaptive and self-learning intelligent systems (RAI)

12. Expected Outputs:

- IoT System for data acquisition
- AI-enabled irrigation advisory platform.
- Farmer dashboard.
- Multi-source integrated agricultural data repository.
- Real-time irrigation recommendation system.
- Forecasting and alert mechanisms.
- LLM-based multilingual advisory module.



Fig. Software Output Sample

13. Modern AI Governance and Paradigms:

Area	Relevance	Recommended Consideration
Privacy & Data Protection	High	Ensure secure handling of farmer information, plot boundaries, sensor data, and soil reports through role-based access control, encryption, secure APIs, and data anonymization wherever required. Compliance with Digital Personal Data Protection (DPDP) Act and institutional data policies should be ensured.
AI Security & Trustworthiness	High	Protect AI models and IoT infrastructure against cyber threats, unauthorized access, data tampering, and adversarial attacks. Implement secure communication protocols, model validation mechanisms, audit logs, and periodic security assessments to maintain reliability and trust in advisory recommendations.
Responsible AI	High	Ensure transparency, fairness, explainability, and accountability in AI-generated irrigation advisories. Models should provide interpretable recommendations and avoid bias arising from variations in soil types, crop varieties, or geographic regions. Continuous monitoring and validation should be carried out before large-scale deployment.
Generative AI / LLMs	High	Utilize Large Language Models (LLMs) to convert technical predictions into farmer-friendly multilingual advisories in English, Kannada, Marathi, and Hindi. Examples include recommendations such as “Irrigate tomorrow morning for 2.5 hours” or “Rainfall is expected; irrigation is not required today.”

		LLMs can also support chatbot-based query handling and knowledge dissemination.
Agentic AI	Medium to High	Intelligent AI agents may autonomously analyze sensor data, weather forecasts, crop stage, and electricity availability to generate personalized irrigation schedules, fertigation recommendations, and alert notifications. Agentic workflows can support proactive decision-making and continuous monitoring across thousands of farms.
Human-in-the-Loop Decision Making	High	AI recommendations should complement, not replace, agronomists, field supervisors, and farmers. Critical decisions regarding irrigation, fertigation, and stress management should incorporate expert validation and farmer feedback. Mechanisms for manual override and advisory acceptance should be provided within the dashboard and mobile application.
Regulatory / Governance Considerations	Medium to High	Ensure adherence to national policies and standards related to AI, geospatial data, IoT deployment, and digital agriculture. Establish governance mechanisms for data ownership, consent management, data sharing, model versioning, and periodic audits. Clearly define responsibilities among KIAAR, GBL, technology providers, and farmers to ensure ethical and sustainable use of AI technologies.

Part B: Program-wise AI Use Case Mapping

14. Programs Mapped:

Institute: KJSIT		**Institute: KJSSE**	
Information Technology	√	Information Technology	√
Computer Engineering	√	Computer Engineering	√
Artificial Intelligence & Data Science	√	Artificial Intelligence & Data Science	√
Electronics and Telecommunication	√	Electronics & Telecommunication Engineering	√
		Computer & Communication Engineering	√
		Electronics & Computer Engineering	√
		Electronics Engineering (VLSI Design & Technology)	×
		Computer Science & Business Systems	√
		Mechanical Engineering	×
		Robotics & Artificial Intelligence	√

15. Technology Domains Mapped:

Program	**Technology Domain**	**Layer**	**Role in Use Case**	**Inter-Disciplinary Dependencies**
CE / IT (Primary)	Computing Foundations	DBMS	Designing Database for all data sources	Receives sensor data from KJSSE, Farmer data from KIAAR/GBL
CE / IT (Primary)	Programming and Software Development	Software	Development of Dashboard for Data Visualization/ Mobile app	Receives sensor data from KJSSE, Farmer data from KIAAR/GBL

CE / IT (Primary)	Artificial Intelligence	AI Models	Trend analysis, Predictions, Recommendations, Forecasting	Receives sensor data from KJSSE, Farmer data from KIAAR/GBL
ETRX (KJSSE)	Networks and Security	IoT	Acquisition of soil/weather etc. data	-
	Emerging Technologies	Edge Computing/ Cloud Computing		-

Part C: Course-level AI Use Case Mapping

16. Course-level AI Use Case Mapping:

Institute: KJSSE **Program:** Electronics & Computer Engineering (EXCP)			
Course	**Contribution to Use Case**	**Relevant Topics from Syllabus**	**Potential Modes of Integration**
Analog Electronic Circuits (Sem III)	Analog front-end for soil moisture, temperature, and rainfall sensor interfacing	Op-amp conditioning, capacitive sensor interfacing, ADC biasing	Lab: design signal conditioning circuit for capacitive soil moisture sensor
Python Programming Laboratory (Sem III)	Weather API integration, data parsing, basic ML scripting for irrigation models	REST API calls, JSON parsing, pandas, matplotlib, data visualisation	Lab: fetch weather forecast from open API, parse JSON, plot temperature and rainfall
Networks, Signals and Systems (Sem III)	Signal processing for multi-sensor agricultural data streams	Sampling, convolution, frequency analysis, noise in sensor signals	Tutorial: analyse soil moisture sensor output signal; design sampling strategy
Data Structures and Algorithms (Sem III)	Efficient spatial data management for geofenced plots of 25,000 farms	Spatial data structures (quad-tree, k-d tree), indexing, priority queues	Assignment: design spatial lookup structure for plot-level sensor data retrieval
Microprocessor and Microcontroller (Sem IV)	IoT sensor node firmware for soil and weather data acquisition and wireless upload	GPIO, ADC, SPI/I2C, LoRa module interfacing, low-power sleep modes	Lab: program microcontroller sensor node for soil moisture and temperature logging
Analog and Digital Communication (Sem IV)	Wireless transmission from field sensor nodes to cloud IoT gateway	LPWAN, LoRa, wireless protocol stack, data framing, range planning	Case study: design LoRa-based communication architecture for distributed field sensors
Database Management Systems (Sem IV)	Multi-source agricultural database integrating soil, weather, crop, and farmer data	ER design, spatial databases, SQL, data integration, normalisation	Mini-project: design integrated agricultural DB schema for irrigation advisory system
Web Programming	Farmer dashboard web interface for	HTML/CSS, JavaScript, chart.js, REST API	Lab: build simple farmer-facing dashboard with soil

Laboratory (Sem IV)	displaying irrigation recommendations	consumption, map APIs	moisture and advisory display
Sensors and Measuring Instruments (Sem V)	Soil moisture, weather station, tensiometer, and rain gauge sensor deployment	Sensor selection, calibration, accuracy analysis, IoT field deployment	Lab: deploy and calibrate soil moisture sensor; assess accuracy over soil types
Automation and Control System (Sem V)	Automated pump scheduling based on AI advisory and electricity availability	PLC control, relay control, scheduled automation, SCADA basics	Lab: design automated irrigation pump controller with soil moisture threshold trigger
Advanced Python Programming and Applications (Sem V)	ML model pipeline: feature engineering, model training, Flask API for advisory serving	scikit-learn, feature engineering, Flask/FastAPI, model deployment	Mini-project: build prediction pipeline for next irrigation date from multi-sensor data
Software Engineering (Sem V)	Architecture design for multi-component advisory platform and API specifications	SDLC, component architecture, API design, UML, testing plans	Assignment: design modular system architecture for irrigation advisory platform
Digital Signal & Image Processing (Sem VI)	Satellite multispectral image processing for NDVI and crop water stress detection	Band analysis, NDVI computation, image segmentation, remote sensing basics	Lab: process satellite image to compute NDVI map and identify crop stress patches
Applied Data Science (Sem VI)	AI models for yield prediction, water stress probability, and irrigation scheduling	ML pipelines, regression, classification, ensemble methods, evaluation	Course project: train yield prediction model from soil, weather, and crop stage data
Computer Communication Networks (Sem VI)	IoT gateway cloud connectivity, MQTT pipelines, and API integration	MQTT, REST, cloud services, IoT security, network architecture	Lab: implement end-to-end MQTT pipeline from sensor to cloud advisory engine
Mobile App Development (Sem VI)	Farmer-friendly mobile app for receiving plot-specific irrigation advisories	Flutter, push notifications, GPS location, map visualisation, offline mode	Mini-project: build Flutter prototype showing plot map with irrigation advisory overlay

Institute: KJSSE **Program:** Computer Science & Business Systems (CSBS)			
Course	**Contribution to Use Case**	**Relevant Topics from Syllabus**	**Potential Modes of Integration**
Artificial Intelligence and Machine Learning (Sem III)	AI/ML model development for irrigation scheduling, water stress, and yield prediction	Supervised learning, decision trees, regression, ensemble methods	Assignment: build and evaluate irrigation date prediction model from historical farm data
Statistical Methods and Probability (Sem III)	Statistical modelling of weather uncertainty and crop water	Bayesian inference, forecasting, confidence intervals, uncertainty quantification	Tutorial: incorporate rainfall forecast uncertainty into irrigation decision using Bayesian reasoning

	requirement variability		
Data Structures and Algorithms (Sem III)	Spatial data management for 25,000 geofenced sugarcane plots	Quad-tree, k-d tree, priority queues, spatial indexing	Assignment: design data structure for efficient spatial lookup of 25,000 farm plots
Computer Organization and Architecture (Sem IV)	Understanding compute infrastructure for scalable advisory serving to 25,000 farms	Cloud computing architecture, distributed systems, scalability	Discussion: analyse compute and storage needs for serving real-time advisories at farm scale
Operations Research (Sem IV)	Optimisation of irrigation scheduling considering electricity availability and water supply	LP, scheduling algorithms, resource optimisation, constraint-based models	Mini-project: formulate optimal irrigation scheduling as an OR problem
Business Communication and Value Science (Sem IV)	Stakeholder communication strategy for farmer adoption of AI advisory system	Value communication, stakeholder analysis, change management	Assignment: design communication strategy for farmer adoption including trust-building plan
Deep Learning & Gen AI (Sem V)	LLM-based multilingual advisory generation; deep learning for crop yield models	LLMs, multilingual NLP, seq2seq models, deep regression for yield	Course project: build multilingual advisory generation pipeline using LLM for 4 languages
Data Mining and Analytics (Sem V)	Mining multi-year irrigation and yield data for correlation patterns and anomalies	Association mining, temporal pattern analysis, anomaly detection	Lab: mine irrigation dataset for correlations between soil moisture patterns and yield loss
Natural Language Processing (Sem VI)	Multilingual advisory generation in English, Kannada, Marathi, and Hindi	Text generation, machine translation, language detection, multilingual NLP	Mini-project: implement NLP pipeline generating farmer advisory in regional language
Introduction to IoT (Sem VI)	IoT sensor network architecture for soil and weather data collection at field scale	LoRa/NB-IoT, sensor management, MQTT, edge processing	Lab: design IoT architecture for distributed soil sensor network across farm plots
Software Engineering using UML (Sem VI)	System design documentation for multi-component advisory platform	Use case modelling, component diagrams, API specifications, V-model testing	Assignment: create full UML model for the irrigation advisory system components
IT Project	Project	Project lifecycle, risk	Assignment: develop phased

Management (Sem VI)	management for phased large-scale deployment across 25,000 farms	management, agile methodology, resource planning	rollout project plan covering pilot to full deployment

Part D: Inter-disciplinary Faculty Committee

17. Faculty Owner(s):

- Dr. Shyamal Virnodkar, Professor, Department of Computer Engineering, K J Somaiya Institute of Technology **(PI)**
- Dr. Radhika Kotecha, Professor, Department of Information Technology, K J Somaiya Institute of Technology
- Dr. Ninad Mehendale, Associate Professor, Department of Electronics Engineering, K J Somaiya School of Engineering

Team:

- Dr. Umesh Shinde, Associate Professor, Department of Basic Science and Humanities, K J Somaiya Institute of Technology
- Dr. Dinesh Auti, Assistant Professor, Department of Electronics Engineering, K J Somaiya School of Engineering
- Dr. Balwant Singh, Assistant Professor, Department of Electronics Engineering, K J Somaiya School of Engineering

18. Program-level Integration Coordinators:

Institute	Program	Coordinator
KJSIT	e.g. IT	Faculty Name & Designation
KJSSE	e.g. Mechanical	Faculty Name & Designation

19. Course-wise Integration Coordinators:

Institute	Program	Course	Course In-charge
KJSIT	e.g. IT		Faculty Name & Designation
KJSSE	e.g. Mechanical		Faculty Name & Designation

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